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Digitization of healthcare organizations: The digital health landscape and information theory

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ABSTRACT

Background: This paper builds on an integrated framework of digital opportunities, based on information theory that was initially proposed by Claude Shannon: 1) providers of infrastructure for transmitting information; 2) platforms, devices, and software to bundle, store and deliver others' content; 3) original content creator); 4) platforms enabling peer-to-peer exchange; and 5) connected intelligence embed in devices. We aimed to clarify the meaning of digitization of organisations in the healthcare landscape and answer 'how can healthcare organisations compete in a world that is rapidly digitizing?'.

Methods: To gain more insight into the views of different stakeholders, we invited 12 participants with a wide range of backgrounds to a workshop. Stakeholders included people representing the views from patients, carers, local hospitals, pharmacy retailers, health insurers, health services researchers, engineers, and technology and pharmaceutical companies. We organised the workshop at the IMD business school in Lausanne and used Switzerland as a case study. We asked participants to take part in group work to identify organisations with digital health products and their key success factors. We evaluated organisations' aims and competencies, and through which pathway they competed.

Main findings: There was overlap in some key success factors, such as security, interoperability, low cost, and enough people using it, between different types of digital opportunities of the framework. No organisations that were identified during the workshop were currently operating in different opportunities. We found a range of different ways through which digital health organisations partnered with other organisations. The relatively smaller organisations identified in our workshop operated through one pathway.

Principal conclusions: This work has implications for organisations in understanding the digital health landscape and their competitors' digital initiatives. Firstly, organisations can identify where they operate and map strengths and limitations of their digital initiatives. Secondly, they can learn from different competition pathways: 1) develop new capabilities faster; 2) collaborate with technology organisations; 3) create collaborations with traditional competitors; and 4) create a new business model. This knowledge needs to be the basis for the development of digital solutions, which will be crucial for the survival of many organisations in the healthcare landscape.

1. Introduction

1.1. Background

Almost three decades ago, IBM nearly did not survive when it failed to adapt to the growing need for personalised rather than centralised computers. After recovery, IBM again struggles to keep up with the changing landscape of digital health and, so far, for example with its Artificial Intelligence branch, Watson, to healthcare issues [1]. This example illustrates challenges for traditional healthcare and medical device companies to digitize their products and services and compete against many digital health start-ups and other non-traditional healthcare companies [2].

Traditional healthcare organisations have realized that they need to change the way they work, for example, over the two years most of the large pharmaceutical companies have started to employ a chief digital officer [3]. Still there are large uncertainties about how to digitize processes and what the different options to leverage their potential are. There are differences in culture and approaches to development and manufacturing between traditional healthcare organisations and start-

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ups. Whilst pharmaceutical companies are used to high-cost and lengthy research and development for drugs, they adopt a far stronger risk-averse approach for engaging with digital health tools that have not shown a clear return on investment.

One issue for all organisations is the challenge to successfully innovate in healthcare. Digitization has revolutionised society in a way we could not have imaged in the 20th century and has the potential to disrupt healthcare and deal with long-standing issues of quality, spiralling costs, rewarding value [4]. However, there are many promising digital health innovations without widespread adoption and many digital health interventions are abandoned when people fail to scale them up or sustain use over time at an organisation or system level [5,6].

The factors influencing non-adoption and abandonment are complex and include: the health condition, technology, value proposition, adopter system (comprising professional staff, patient, and lay caregivers), organization(s), wider (institutional and societal) context, and interaction and mutual adaptation between all these factors over time [5,7]. Some specific reasons are the complexities related to regulation [8,9] and the conflicting interest of different stakeholders in the healthcare ecosystem [10,11]. A typical example of a regulation challenge is that for an innovation to be approved by regulatory authorities could take years, which is in stark contrast to the rapid development and change of digital solutions. An example of the second reason is the lack of incentives for healthcare systems such as the US to reduce the length of recovery of patients and number of services provided, since the hospital earns revenue by night of stay (like hotels) and physicians earn fees for services [12].

1.2. Aim and theoretical framework

The aim of this paper is to clarify the meaning of digitization of organisations working in the healthcare landscape and to find out from stakeholders in the healthcare ecosystem what would be required to increase sustainable adoption of digital solutions. This paper builds on an integrated framework of digital opportunities proposed by Shervani and Challagalla [13] based on information theory, which was initially proposed by Claude Shannon [14]. This framework helps organisations to understand the digital health landscape and their competitors' digital initiatives. Also, the framework can be used to identify opportunities to compete in the digital health landscape.

We choose this framework after reviewing the medical and management and organisation theory literature, including systematic reviews on frameworks and theories. Many of these frameworks focus on strategies for product development within an organisation, whilst we were more interested with our research question to research how organisations can compete against each other in the healthcare landscape for which the Shervani and Challagalla framework seemed most appropriate. Information theory is relevant to the digital space because Shannon defined previously amorphous concepts such as "data," "information," and "knowledge." Information has a special meaning in Shannon's theory – he equated it with uncertainty or surprise. Also, Shannon was the first to describe a general communication system that can be applied to any form of information transmission (speaking, writing, music, computer code, digital signals between devices, etc.) through any type of medium (wire, air, fiber optics). Shannon illustrated that every communication system contains an information source, a transmitter, a channel or pipeline, a receiver and a destination.

Shervani and Challagalla outline four pathways through which organisations can address the challenges related to making changes: 1) develop new capabilities faster on its own compared to competitors; 2) collaborate with technology organisations to access to their research and development facilities and capabilities; 3) create collaborations with traditional competitors and share investments and research and development; and 4) create a new business model. These pathways are not mutually exclusive, and organisations can use more than one or all pathways at the same time. The main question that is asked is 'how can traditional organisations compete against newcomers in a world that is rapidly digitizing?'.

2. Materials and methods

To identify different organisations in the healthcare landscape and stakeholders' perceptions on barriers and facilitators towards the successful digital health innovations, we organised a workshop on 25th August 2017 at the IMD business school in Lausanne. Given the background and knowledge of our participants, we used Switzerland as case study. Switzerland scores consistently high on measurements for information and communication technologies, innovation, and healthcare system performance, but spends a disproportionally high percentage on healthcare compared to other high-income countries [15].

To gain more insight into the views of different organisations, 12 participants with a wide range of backgrounds attended the workshop, including people representing the views from patients, carers, local hospitals, pharmacy retailers, health insurers, health services researchers, engineers, and technology and pharmaceutical companies. We asked participants to take part in a group work session in which we used the widely adopted Shannon's Model of Communication, also referred to as information theory that incorporates the information source, transmitter, channel, receiver, and destination (Fig. 1).

This framework was more recently used by Shervani and Challagalla to classify digital opportunities with the following five 'Shannon Stacks' (Fig. 2): 1) providers of infrastructure as channels ('information pipes') for transmitting information (e.g. telecommunication providers); 2) platforms, devices, and software to bundle, store and deliver others' content/products as transmitters (e.g. integrated software companies such as Epic Systems Corporation); 3) original content creators as information sources (e.g. personalized medicine company Assurex Health, Qiagen, or R&D departments in corporations and universities); 4) platforms enabling peer-to-peer exchange to uncover 'private' information and assets as receivers (e.g. patient network websites such as patientslikeme.com); and 5) distributed, connected intelligence embed in devices and entities (Internet of Things etc.) as destinations (e.g. tracker devices such as Fitbit). Participants were asked to identify Swiss

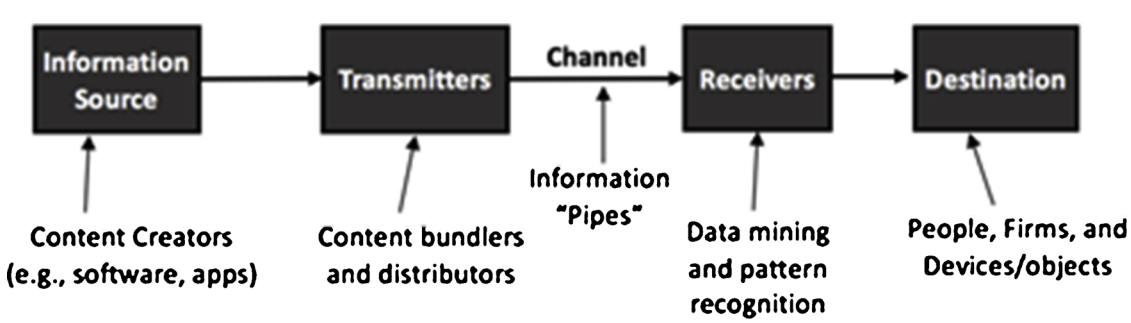

Fig. 1. Shannon's Model of Communication and Players [14].

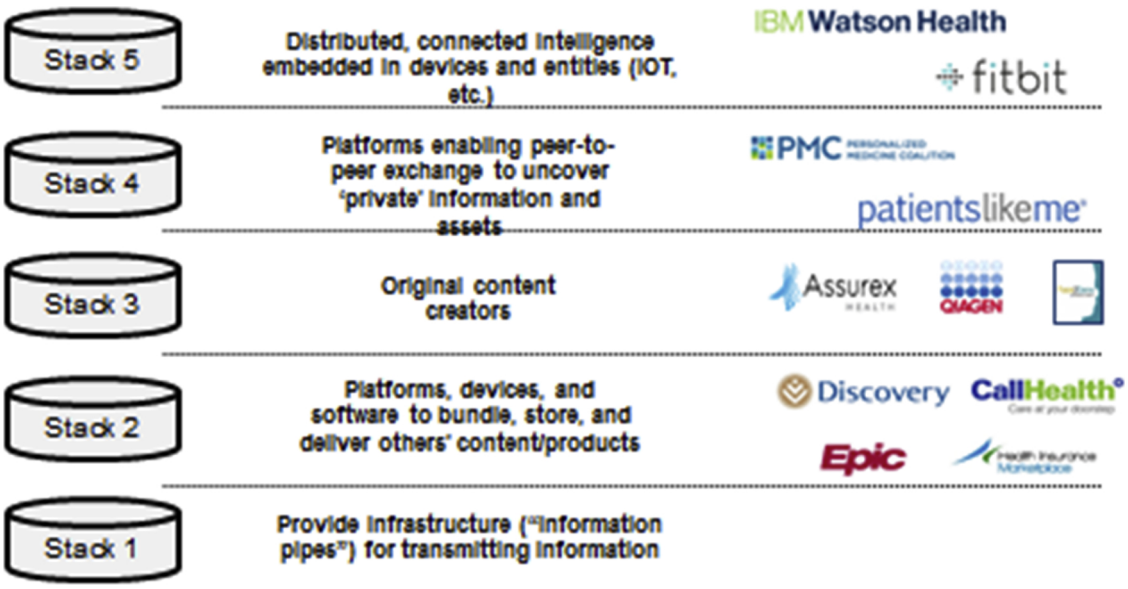

Fig. 2. Shannon Stacks applied to the life sciences [13].

digital health organisations and key success factors for the five different Shannon Stacks.

After the workshop we used Google to search the Internet on 4th September 2017 to collected information on: 1) aims, values and competencies mentioned on the organisations' websites; and 2) partnerships used by the organisations identified during the workshop. We collected information using a structured data extraction sheet. We compared the success factors mentioned in the workshop with the aims, values/competencies of the organisation's website. We identified the main goals of collaborations and through which of the four pathways organisations competed: 1) develop new capabilities faster; 2) collaborate with technology companies; 3) create collaborations with traditional competitors; and 4) create a new business model.

3. Findings

3.1. Key success factors

Table 1 provides an overview of digital health organisations in Switzerland that were mentioned for the five Shannon Stacks. There was overlap in some key success factors between the five different Shannon Stacks such as security, interoperability, low cost, and a sufficient number of people using the digital health innovation. Other key success factors seemed more specific for particular Shannon Stacks, such as fun innovations for Internet of Things innovations in stack 5, personalisation for peer-to-peer platforms in stack 4, validation for original content in stack 3, no vendor lock-in for bundle/storage platforms in Stack 2, and reliability for information pipes in stack 1. Digital health organisations in Switzerland seemed to operate in one Shannon Stack only.

We describe the aim, tools and values of specific digital health organisations for each stack in the Appendix (Table A1–A5). When comparing the values of organisations with key success factors, we did not find much overlap apart from security and integration/interoperability of services. For Shannon Stack 5, fun and appealing were mentioned as factors in the workshop and the company 'Care Servant' states on their website that they provide entertainment and great user experience. Consent was mentioned during the workshop and by the Swiss Biobank platform for Shannon Stack 4. For stack 2, ease of use and accessibility was mentioned during the workshop. Swisscom mentions fast web development on their website, which was also mentioned during the workshop for Shannon Stack 1.

3.2. Pathways

Table 2 shows a mix of different pathways through which organisations in Switzerland digitize their services, being: 1) develop new capabilities faster; 2) collaborate with technology companies; 3) collaborating with competitors; and 4) create a new business model. For pathway 1, organisations such as the Swiss Biobank Platform, Swiss Institute of Bioinformatics and Swiss Ethics are national non-profit organisations who work together with Swiss universities and hospitals to increase the reach of their services more quickly. Also, companies such as Versicherix P2P insurance works with an incubator to speed up the set-up of their services while Med Gate partners with physicians, hospitals and pharmacies in the US. For pathway 2, collaborations between healthcare and IT companies were seen for 1 drop diagnostics, Galenica and Medbase. For pathway 3, Domo safety and CareServant worked with competitor IT companies. For pathway 4, Midata.coop and Post Health attempted to create new business models (Table 2).

4. Discussion

4.1. Summary

There are major shifts in the healthcare industry related to the digitization of patients, healthcare clinics, devices, and medication [16]. Non-traditional healthcare companies such as Google and Amazon are moving into the healthcare space and thereby creating tension. For example, Google's parent company Alphabet is hiring top engineers for its life sciences arm Verily and Amazon might be selling prescription drugs soon [17]. This article addressed Shervani and Challagalla's [13] assertion that whether traditional organisations can better compete against newcomers depends on who learns others' businesses faster. Our research question addressed how organisations can compete against each other in the healthcare landscape for which based on a review of the literature the Shervani and Challagalla framework seemed most appropriate. We organized a workshop with the aim to contribute to clarify the meaning of digitization of organisations working in the healthcare landscape. Strengths of our work include incorporating a widely used theory and theoretical framework for our study and inviting stakeholders from different background to provide their views. Limitations include our focus on one country only, Switzerland, which limits generalisability of the results. However, by drawing from organisations in other settings we provide a more general overview of the

Table 1
Shannon Stacks examples of Swiss digital health organisations and key success factors.

Shannon Stacks	Examples of Swiss organisations	Key success factors mentioned[a]
Stack 5: distributed, connected intelligence embed in devices and entities	5.a. Domo safety, smart home care services 5.b Care servant, hospital software and hardware 5.c 1 Drop Diagnostics, biomarker measurement device	- Security - Sufficient number of users - Regulatory compliance - Low cost - Evidence of benefits - Clear target market - Utility - Appealing - - Fun
Stack 4: platforms enabling peer-to-peer exchange to uncover 'private' information and assets	4.a Midata.coop, platform for citizens to gather health-related data 4.b Swiss Biobank platform, national biobank network 4.c Versicherix P2P insurance, peer-to-peer insurance company	- Security - Sufficient number of users - Interoperability - Dynamic consent of patients - Personalised - Interface neutrality - Reproducibility
Stack 3: original content creators	3.a Swiss Institute of Bioinformatics (SIB), bioinformatics foundation 3.b Swiss ethics, joint working group for ethics committees 3.c Galenica (KompendiumApp and online version), pharmacy network	- Low cost - Accessibility - Purpose of the data - Validation - Credibility - Language - Relevance - Accuracy of information
Stack 2: platforms, devices, and software to bundle, store and deliver others' content/products	2.a Medbase (Migros), wellness services for companies 2.b Med Gate, provider of telemedicine services	- Security - Interoperability - Searchability - Ease of use - No vendor lock-in - Interface neutrality - Good user-interface - Capacity
Stack 1: provide infrastructure ('information pipes') for transmitting information	1.a Swiss Post, eHealth platform 1.b Swisscom, telecoms company	- Security - Interoperability - Regulatory compliance - Ownership - Fast - Reliable (uninterrupted connectivity)

[a] overlapping factors in italic.

current situation below.

The first pathway is for a company to develop new capabilities faster on its own compared to competitors such as Versicherix P2P insurance works with an incubator to speed up the set-up of their services while Med Gate partners with physicians, hospitals and pharmacies in the US. IBM did this by developing its Artificial Intelligence platform and calling it after its first CEO, Thomas Watson, whereby, IBM hoped to develop Artificial Intelligence quicker than their competitors. Another example is Mediq, a Dutch healthcare company bought by Brocacef in 2014 providing pharmaceuticals and medical supplies by operating apothecaries and delivering supplies to medical centres and patients. While the company was making a loss in 2012, through strategic investment in their revenue-generating delivery of supplies to medical centres and patients they managed to learn quicker than their competition to improve their services. They successfully gained market share by implementing a new planning system, Slimstock, to optimise refill of their 250 pharmacies across the Netherlands. The benefit of this system is that inventory is centrally managed, and pharmacists have to make replenishment orders themselves. As a result, there was a 20% decrease in overall stock levels and 4% service level improvement. This helped Mediq to be the market leader in the Netherlands and gain success in 14 other countries [18].

The second pathway is for a company to collaborate with technology companies to access to their research and development facilities and capabilities, such as for 1 drop diagnostics, Galenica and Medbase in Switzerland. This appears to be a frequently used collaboration [19,20]. For example, pharmaceutical company Otsuka, creator of the psychiatry drug Abilify, jointly developed a digital ingestion tracking system with medical device company Proteus Digital Health, which developed the sensor. In November 2017, Abilify MyCite became the first Food and Drug Administration approved digital pill in the US [21]. Pharma company AstraZeneca signed an agreement with PatientsLikeMe, a patient network that improves lives and a real-time research platform that advances medicine. PatientsLikeMe will provide AstraZenea with access to its global network to support its patient-driven research objectives [22]. Also, Johnson and Johnson has used this approach with IBM and Google's Alphabet [23]. IBM Watson collaborated with electronic health records company Epic Systems to provide AI and more comprehensive clinical decision support capabilities to electronic health records. IBM also used its ability to analyse large amount of data by collaborating with medical device company Medtronic [24].

The third pathway is for a company to create collaborations with traditional competitors and share investments and research and development, such as Domo safety and CareServant in Switzerland. The life sciences industry has relied on this form of collaboration for a long time, even though it might sound unusual. For example, Qiagen, provider of molecular diagnostics, and Centogene, a genetic diagnostics company, announced a collaboration to broaden the rare and genetic disease products available for both companies' customers [25]. However, in the digital health landscape this strategy is still less common.

The fourth pathway is to be a rule maker or breaker and create a new business model like Midata.coop and Post Health in Switzerland.

Table 2
Shannon Stacks examples of Swiss digital health organisations and collaboration pathways.

Shannon Stack number	Organisation	Collaborator/partner	Web link	Pathway	Goal of collaboration
5.a	Domo safety	Qloudlab, medical device company for point of care testing	http://qloudlab.com/new-collaboration/	3	Partner on one development project
5.b	CareServant	VAPC Interactive Services, Austrian IT company	https://www.vapc.at/en/	3	Together run over 10,000 bed side terminals across Europe, engaging more than 500,000 patients every year
5.c	1 drop diagnostics	Massachusetts General Hospital (MGH)	http://www.massgeneral.org/	2	Obtain clinical guidance, specific identification of unmet clinical needs, and the environment to perform validation studies from clinicians at MGH.
4.a	Midata.coop	Universities in Switzerland, and internationally Charité and Berlin Institute of Health, Medical Delta, TNO, INDEPTH-Network.org (in Ethiopia and Vietnam)	https://midata.coop/	4	Enables citizens to securely store, manage and control access to their personal data by helping them to establish and own national/regional not-for-profit MIDATA cooperatives
4.b	Swiss Biobank platform	University Hospitals	http://www.swissbiobanking.ch/	1	Non-profit organizations officially commissioned to conduct biobanking activities.
4.c	Versicherix P2P insurance	Swiss Cubator	http://www.swisscubator.com/	1	Accelerating and incubating Swiss insurtech and eHealth startups.
3.a	Swiss Institute of Bioinformatics (SIB)	Swiss Universities and hospitals, industrial and international partners	https://www.sib.swiss/about-us	1	Sharing its expertise in storage, analysis and dissemination of large biological datasets and through training and collaborations with research institutes, industrial and international partners
3.b	Swiss ethics	Swiss Universities and hospitals	http://www.swissethics.ch/eks_e.html	1	Coordination and standardisation of working processes, education and training of the members of the ethics committees
3.c	Galenica (KompendiumApp and online version)	Documed, publishing house, and HCI Solutions AG, health software	https://itunes.apple.com/ch/app/documed-kompendium/id418131146?l=en	2	Launch a digital version of the Swiss Drug Compendium for iPhone, iPad and iPod touch (KompendiumApp) as well as an optimized online version for other smartphones, e-book readers, Windows and Mac.
2.a	Medbase	Migros (supermarket chain) fitnessparks	https://www.medbase.ch/ueber-medbase/portrait-medbase/	2	Offer intervention support preventive measures such as fitness training or wellness.
2.b	Med Gate	Swiss hospitals and health centres	https://www.medgate.ch/en-us/partnernetwork	1	The Medgate Partner Network consists of general physicians, specialists, hospitals, and pharmacists networked with Medgate.
1.a	Post Health	Swiss hospitals and health centres	https://www.post.ch/en/business/a-z-of-subjects/industry-solutions/industry-solution-healthcare	4	Consolidating numerous niches into a comprehensive e-health solution that meets the needs of patients, doctors, consultants and the public sector.
1.b	Swisscom	NA	NA	NA	NA

New York healthcare insurance company Oscar collaborated with Apple's Misfit wearable device in 2014. People with Oscar insurance could opt to receive a free Misfit wristband pedometer that connects automatically to Oscar's app and pays people to exercise. This was aimed to create a new business model and reinvent the healthcare insurance industry, which has growth challenges [26]. While Oscar does not use the data to inform its health plans, it uses it to engage better with its customers and found that 80% of customers who download the app track their steps [27]. Another example is the company Betterdoc in Germany that recruited 2500 leading experts in their respective fields to advise patients as to which specialist(s) can best provide care tailored to their condition. They found that 60% of planned operations are unnecessary and can be avoided by second opinion; 86% of heavy complications can be avoided when the right specialist is performing the surgery. In this low-cost, scalable and novel model, patients receive these services for free and their employer or insurance company pays [28].

4.2. Conclusions

Our results have implications for organisations in understanding the digital health landscape and their competitors' digital initiatives. Firstly, organisations can identify in which Shannon Stack they operate and map strengths and limitations relevant to their digital health innovations. In Switzerland, the digital health organisations that we identified seemed to operate in one Shannon Stack only. While there were specific success factors for each stack, overlapping success factors were security, interoperability, low cost, and a sufficient number of people using the digital health innovation.

Secondly, organisations can learn from the four pathways through which organisations can compete to develop digital health innovations. There are benefits and downsides for different four pathways by which organisations can compete to develop digital health innovations and organisations need to research their strengths, weaknesses, opportunities and threats before choosing a pathway to compete. The organisations identified in our workshop operated through one of the four different pathways whilst large companies, such as IBM, appear to attempt different pathways. Challenges to achieve a better competitive position through one or more of these pathways for healthcare organisations include finding new ways to distance collaborator organisations and limit the responsibility of a traditional company. An understanding of the roles and incentives of the other stakeholders is required, for which collaboration is key. Organisations need to undertake research to understand others and use this knowledge as a basis for the development of digital solutions, which will be crucial for the survival of many organisations in the digital health landscape.

Conflicts of interest

Michelle van Velthoven is director of Dutches Consulting Ltd and provides advice to clients in the life sciences.

Author's contributions

Michelle van Velthoven wrote this paper and Carlos Cordon revised the paper. Also Carlos Cordon and Goutam Challagalla provided important intellectual comments.

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Appendix A

Table A1
Stack 5 example Swiss digital health organizations.

Organisation	Website	Aim (copied from website)	Tools/platform/content	Values/competencies (copied from website)
Domo safety	http://www.domo-safety.com/	Provider of behaviour-based and medical data analytics in real-time to deliver preventive and proactive services to health professionals	Emergency button, activity sensors for in the house, analytics	- Cloud-based & highly secured data transfer platform, fully compliant with European regulations - Validated health professionals interface - Integration with call centers - Connectivity with homecare sensors - Compliant interface with existing e-health tools - Integration expertise at both end - Data collection with hardware partners
Care Servant	http://www.careservant.com/en/	Patient engagement that is pleasant, personalized and profitable; Combine software and content with innovative hardware into one complete solution for hospitals	Software (applications), content (multimedia content on entertainment, information and administration) and devices (consumer mobile devices, e.g. iPads)	- Software; relevant applications, true personalisation and great user experience, all centrally managed - Content; multimedia content on entertainment, information and administration, all fully integrated - Devices; innovative use of consumer mobile devices (e.g. iPads) to bring patient experience to a new level
1 drop diagnostics	https://www.1dropdx.com/	Easily measure biomarkers from a tiny drop of blood immediately in front of the patient	1Drop Chip, 1Drop Reader, 1Drop App	- Easy measurement - Quality results; analyse with high-sensitivity, accuracy and precision - Monitor health; detect multiple biomarkers of cardiovascular health - Personalized health insights - Share data with care team and electronic health record

Table A2
Stack 4 example Swiss digital health organizations.

Organisation	Website	Aim (copied from website)	Tools/platform/content	Values/competencies (copied from website)
Midata.coop	https://www.midata.coop/	Enables you to gather all your different health-relevant and other personal data in one secure place. You can decide to share data with friends or physicians or to participate in research by providing access to subsets of your data. In that way you contribute to the development of new treatments for OUR HEALTH.	MIDATA IT Platform.	- Citizen owned. - Not for profit. - Open source. - Transparent. - Secure. - Regional.
Swiss Biobank platform. Founded by Swiss National Science Foundation	http://www.swissbiobanking.ch/	Respond to the increasing requests from biomedical researchers regarding quality and interconnectedness of biobanks for research purposes.	Central web based catalogue of existing and de novo biobanks with data and samples' access policies.	- Reliable customer-oriented network of biobanks. - Traceability, access and distribution of samples and data for research purposes. - General information and consent of citizens. - Support to investigators, collaboration with the 5 University Hospitals. - Biobank sustainability.
Versicherix	http://versicherix.ch/	Peer-to-peer insurance company.	Website.	- Simple functionality. - Choose insurance. - Find friends. - Save money.

Table A3
Stack 3 example Swiss digital health organizations.

Organisation	Website	Aim (copied from website)	Tools/platform/content	Values/competencies (copied from website)
SIB Swiss Institute of Bioinformatics	http://www.sib.swiss/	Independent, non-profit foundation that includes some 65 bioinformatics research and service groups and some 800 scientists from the major Swiss schools of higher education and research institutes. Providing life scientists and clinicians in academia and industry with world-class bioinformatics resources and services, and by leading and coordinating the field of bioinformatics in Switzerland.	150 bioinformatics resources accessible on the SIB web portal www.expasy.org.	Empowering advances in life sciences and health.
Swiss ethics	http://www.swissethics.ch/index_e.html	The Swiss Ethics Committees on research involving humans have formed a joint working group (swissethics).	e-consent.	- coordination and standardisation of working processes - education and training of the members of the ethics committees
Galenica	http://www.galenica.com/en/ueber-galenica/geschichte/geschichte.php#mode:list	Leading fully-integrated healthcare provider in Switzerland. Operates the largest network of pharmacies in Switzerland and offer well-known own brands and products, exclusive brands and products from business partners as well as a variety of on-site health services and tests for customers.	Digital version of the Swiss Drug Compendium for iPhone, iPad and iPod touch (KompendiumApp) as well as an optimized online version for other smartphones, e-book readers, Windows and Mac.	- Helps to shape the Swiss healthcare market of the future through innovation. - Reliable, attractive partner and employer. - Generating reliable, attractive returns with manageable risks.

Table A4
Stack 2 example Swiss digital health organizations.

Organisation	Website	Aim (copied from website)	Tools/platform/content	Values/competencies (copied from website)
Medbase (Migros)	https://www.medbase.ch/medbase/	Innovative healthcare provider. Medbase employs more than 900 people and has worked closely with Migros Fitnessparks for over ten years. The idea: Medical services and intervention support preventive measures such as fitness training or wellness. The Medbase Corporate Health division supports companies with comprehensive health management.	-health report -myChange -MyRelax -Oviva - online nutrition coaching -physio.coach -running.COACH	- 360 ° supervision. - First class medical quality. - Easy access to our services.
Medgate	http://www.medgate.ch/en-us/home	Providing telemedicine services: -Diagnosis with images, Prescribing, Sick-leave certificate, Treatment plan by email or SMS, Dispatch of medicines, Medical hotlines, International Medical Assistance.	Telemedicine center, Medgate App	- Better assessment of skin and eye changes. - Based on clearly defined indications and the most up-to-date scientific knowledge. - Subject to strict guidelines. - Anonymous treatment plan is sent under strict conditions of data security to an email address/cell-phone number of your choice, and only with your express consent. - Established online pharmacies. - Official supplier for the health hotlines of the Federal Office of Public Health. - Guarantees high-quality, and at the same time cost-effective, care abroad.

Table A5
Stack 1 example Swiss digital health organizations.

Organisation	Website	Aim (copied from website)	Tools/platform/content	Values/competencies (copied from website)
Swiss Post	https://www.post.ch/en/business/a-z-of-subjects/industry-solutions/industry-solution-healthcare/vivates-ehealth	The modular design of the e-health platform enables healthcare professionals to compile tailor-made solutions.	e-health platform for electronic patient records, care plan, medication plan, referral, report transfer, quality and surveys.	- Access and exchange patient records. quickly and simply - Access anytime and anywhere. - Simplifying cooperation between healthcare professionals. - Enhancing patient safety. - More efficient and more cost-effective.
Swisscom	https://www.swisscom.ch/en/about/company/portrait/profile.html	Switzerland's leading telecoms company and one of its leading IT companies.	–	- Operational excellence - Transformation - Maximisation of core business. - Growth and fast web development.

Appendix B. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.ijmedinf.2019.01.007.

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